

Asset Classes

One of the base master tables used is Asset Classes. Key attributes are:

- Asset ID
- Asset Class
- Asset Description
- Sub-asset class
- Risk
- Investment Horizon
- Sub-asset description

This data is used in Securities and Portfolio applications to classify and rollup. Some sample data shown below.

AssetID	Asset Class	Asset Description	Sub Asset class	Risk	Investment Horizon	Sub Asset Description
10000	Cash		Cash	Low	Short	Cash balance in Bank account
10001	Equity	Equity asset class	Stock	High	Long	Listed Equity stocks in Exchanges
20001	Mutual Fund	Mutual fund asset class	Stock Funds	High	Long	Stock mutual fund invests principally in equity or stocks
						Fixed-income mutual fund focuses on investments that pay a set rate of return, such as government bonds, corporate bonds, or other debt instruments
20002	Mutual Fund	Mutual fund asset class	Bond Funds	Low	Any	Index Funds invest in stocks that correspond with a major market index such as the S&P 500 or the Dow Jones Industrial Average (DJIA)
20003	Mutual Fund	Mutual fund asset class	Index Funds	Medium	Any	Balanced funds invest in a hybrid of asset classes, whether stocks, bonds, money market instruments, or alternative investments.
20004	Mutual Fund	Mutual fund asset class	Balanced Funds	Medium	Long	The money market consists of safe, risk-free, short-term debt instruments, mostly government Treasury bills.
20005	Mutual Fund	Mutual fund asset class	Money Market Funds	Low	Short	Income funds are named for their purpose: to provide current income on a steady basis. These funds invest primarily in government and high-quality corporate debt, holding these bonds until maturity to provide interest streams
						An international fund, or foreign fund, invests only in assets located outside an investor's home country.
20006	Mutual Fund	Mutual fund asset class	Income Funds	High	Long	Sector funds are targeted strategy funds aimed at specific sectors of the economy, such as financial, technology, or healthcare.
20007	Mutual Fund	Mutual fund asset class	International/Global Funds	High	Long	ETFs are not considered mutual funds but employ strategies consistent with mutual funds
20008	Mutual Fund	Mutual fund asset class	Speciality Funds	High	Long	Tbills are short-term fixed-income securities that mature within one year that do not pay coupon returns.
20009	Mutual Fund	Mutual fund asset class	Exchange Traded Funds (ETFs)	Medium	Any	Treasury notes (T-notes) come in maturities between two and 10 years, pay a fixed interest rate, and are sold in multiples of \$100.
21001	Fixed Income	Fixed Income asset class	T-Bills (Treasury Bills)	Low	Short	Treasury bonds (T-bonds) are similar to the T-note except that it matures in 20 or 30 years.
21002	Fixed Income	Fixed Income asset class	T-Notes (Treasury Notes)	Low	Any	Treasury Inflation-Protected Securities (TIPS) protect investors from inflation. The principal amount of a TIPS bond adjusts with inflation and deflation. Sold as 5, 10 or 30 years maturity.
21003	Fixed Income	Fixed Income asset class	T-Bonds (Treasury Bonds)	Low	Long	Municipal Bond is similar to a Treasury since it is government-issued, except it is issued and backed by a state, municipality, or county
21004	Fixed Income	Fixed Income asset class	TIPS (Treasury Inflation-Protecte	Low	Long	Corporate bonds come in various types, and the price and interest rate offered largely depend on the company's financial stability and its creditworthiness. Bonds with higher credit ratings typically pay lower coupon rates.
21005	Fixed Income	Fixed Income asset class	Municipal Bond	Low	Long	Junk bonds—also called high-yield bonds—are corporate issues that pay a greater coupon due to the higher risk of default.
						A certificate of deposit (CD) is a fixed income vehicle offered by financial institutions with maturities of less than five years.
21006	Fixed Income	Fixed Income asset class	Corporation Bond	Medium	Medium	24 Ct Gold traded in Exchanges
21007	Fixed Income	Fixed Income asset class	Junk Bond	High	Medium	Real estate investment trusts. Typically earn Income through Rental income fpr investments made in Properties.
21008	Fixed Income	Fixed Income asset class	Certificate of Deposit (CD)	Medium	Medium	
31000	Commodities	Commodities	Gold	Low	Long	
41000	REITS	Real Estate Investment Tr	REITS	Medium	Long	

Expose Data as a Service

In order to expose the above data as a service, the following specification can be followed:

- a. Define the endpoint URL for retrieving Asset Class data. For example, you might use the following
URL: /asset-classes/{id}, where {id} is the unique identifier for a specific Asset Class.
- b. Create a REST controller class to handle requests for Asset Class data. This class should be annotated with **@RestController** and should have methods to handle GET requests for retrieving Asset Class data.
- c. Define a method in the controller to handle GET requests for retrieving all Asset Classes. This method should be annotated with **@GetMapping** and should return a list of Asset Class objects. For example:

```
@GetMapping("/asset-classes")
public List<AssetClass> getAllAssetClasses() {
    // code to retrieve all Asset Classes from the database
    List<AssetClass> assetClasses = assetClassService.getAllAssetClasses();
    return assetClasses;
}
```

- d. Define a method in the controller to handle GET requests for retrieving a single Asset Class by ID. This method should be annotated with **@GetMapping** and should accept the ID of the Asset Class as a path variable. For example:

```
@GetMapping("/asset-classes/{id}")
public AssetClass getAssetClassById(@PathVariable Long id) {
    // code to retrieve the Asset Class with the given ID from the database
    AssetClass assetClass = assetClassService.getAssetClassById(id);
    return assetClass;
}
```

- e. Create an Asset Class entity class with the attributes ID, Class Name, Sub-class Name, and Description. For example:

```
public class AssetClass {
    private Long id;
    private String className;
    private String subClassName;
    private String description;

    // constructors, getters, and setters
}
```

- f. Create a service class to handle business logic for retrieving Asset Class data. This class should be annotated with **@Service** and should have methods to retrieve all Asset Classes and a single Asset Class by ID.

- g. Define methods in the service class to retrieve all Asset Classes and a single Asset Class by ID. These methods should call the appropriate methods in a repository class to retrieve data from the database. For example:

```
@Service
public class AssetClassService {
    @Autowired
    private AssetClassRepository assetClassRepository;

    public List<AssetClass> getAllAssetClasses() {
        return assetClassRepository.findAll();
    }

    public AssetClass getAssetClassById(Long id) {
        Optional<AssetClass> assetClassOptional = assetClassRepository.findById(id);
        if (assetClassOptional.isPresent()) {
            return assetClassOptional.get();
        } else {
            throw new RuntimeException("Asset Class not found with ID: " + id);
        }
    }
}
```

- h. Create a repository class to handle database access for Asset Class data. This class should extend the [JpaRepository](#) interface provided by Spring Data JPA and should have methods to retrieve all Asset Classes and a single Asset Class by ID. For example:

```
public interface AssetClassRepository extends JpaRepository<AssetClass, Long> {
}
```

- i. Configure the database connection in your [application.properties](#) file:

```
spring.datasource.url=jdbc:mysql://localhost:3306/your-database-name
spring.datasource.username=your-username
spring.datasource.password=your-password
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
```

- j. Run your application and test the REST API endpoints using a tool like Postman or cURL. You should be able to retrieve a list of all Asset Classes by sending a GET request to /asset-classes, and you should be able to retrieve a single Asset Class by ID by sending an Asset ID.